# I Want to Play a Game

**Halloween · One-sided confidence interval**

A lower bound for mean latch strength

Participation exercise • please show your working

**Jigsaw's message**

*"You advertised that the latches in your company have mean breaking strength at least 50. Many people have bought your products because of your claim. I brought one, which was WEAK! Now, let's play a game. Go pick up 10 latches on the shelf behind you — they are all your products. Then you can measure their strengths using the machine in front of you. If you can show that your claim is true with a 95% confidence level, I will forgive you for selling me a weak latch and let you go. Otherwise, you are a liar — and now you have to pay for it!"*

**Your data.** You randomly pick 10 latches and measure breaking strengths (kg):

$$51,\ 55,\ 48,\ 53,\ 57,\ 50,\ 54,\ 49,\ 56,\ 52.$$

The population standard deviation is *unknown*. Treat the strengths as a random sample from a normal population with mean $\mu$.

**Task.** Give a **95% lower confidence bound** for $\mu$ (write the interval in the form $[L,\ \infty)$). Then decide: does this interval support the advertising claim $\mu \geq 50$ at the 95% level?

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No written submission — a few students will be invited to present in class. Happy Halloween.